KMUTT | SIT — School of Information Technology, King Mongkut's University of Technology Thonburi

"Empowering Boundless Digital Transformation"

CS D-Day 2026

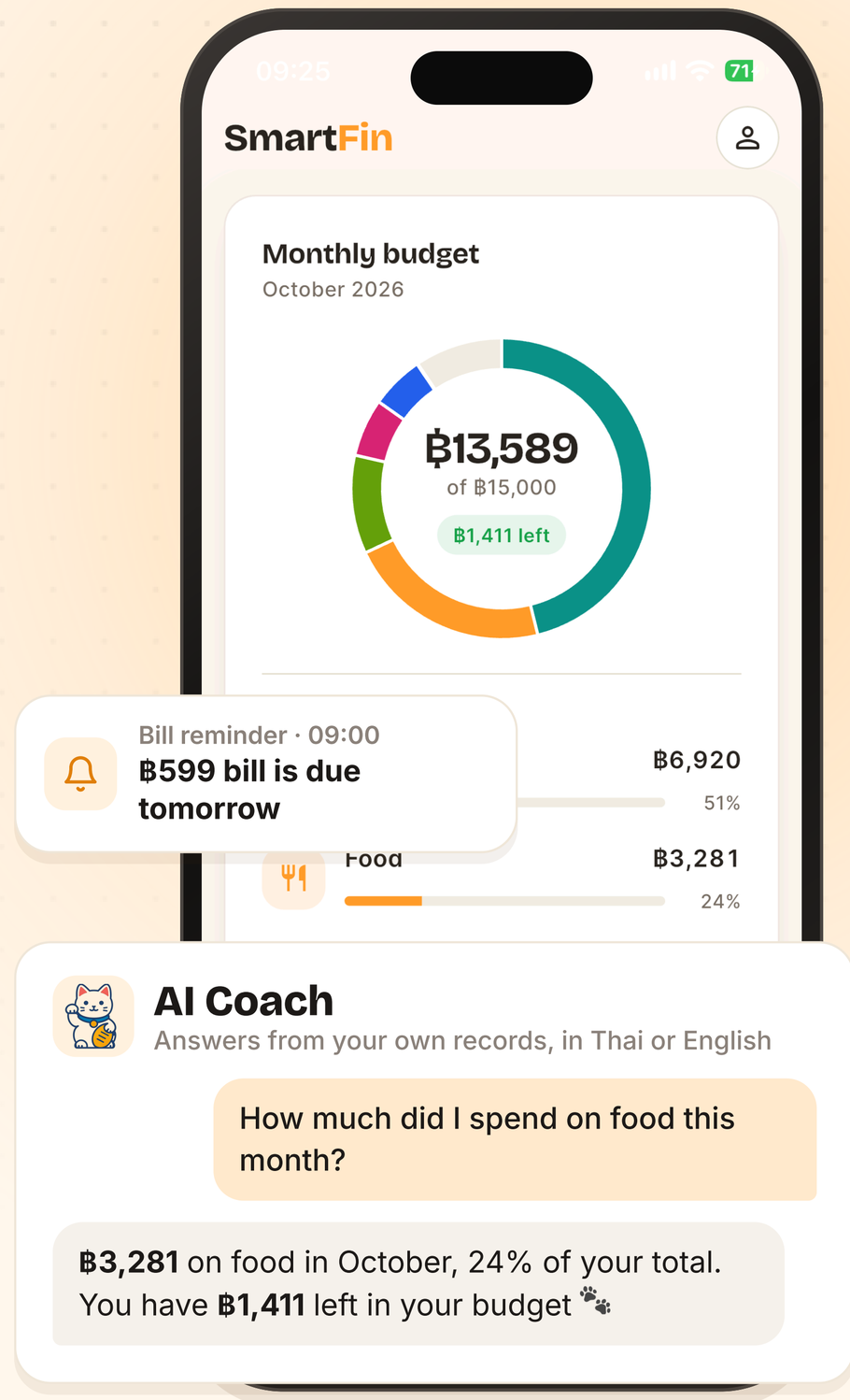
CSC499 Computer Science Capstone Project II

Personal expense tracking app for Thai users

SmartFin

Automatic Expense Tracking with Thai E-Slip Recognition and an AI Financial Coach

Thai payments end in an e-slip, yet budgeting apps still make you **type every expense by hand**. SmartFin reads the slips for you.



Snap the slip, skip the typing

A vision model reads the merchant, the amount in Baht and the date. You confirm before it saves.

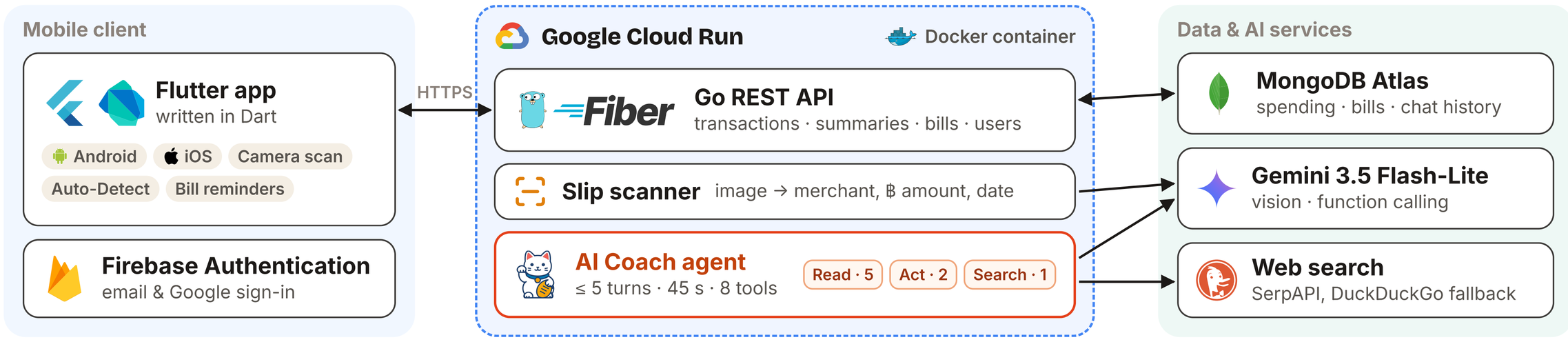
Auto-Detect from the gallery

New e-slips from K PLUS, SCB EASY, Krungthai NEXT and other banking apps are found and offered for review.

Bills that remind you

Subscriptions and recurring bills that never produce a slip, with a reminder the day before they are due.

1 System architecture



2 Tools & techniques

Mobile
- Flutter
- Dart
- photo_manager
- fl_chart

Backend
- Go
- Fiber v2
- Docker
- Google Cloud Run

AI
- Gemini 3.5 Flash-Lite
- Agentic RAG
- Function calling
- Gemini Vision

Data & auth
- MongoDB Atlas
- Firebase Auth
- Google Sign-In
- Web search

Workflow
- GitHub
- Postman
- Figma
- VS Code
- Go & Flutter tests

Project team



Ant Bone Kyaw
66130500802



Hnin Ei Ei Aung
66130500808



Thaw Zin Moe Myint
66130500824

Project advisor



Dr. Debajyoti Pal
Computer Science (English Program)
School of Information Technology
King Mongkut's University of Technology Thonburi

Vote for SmartFin